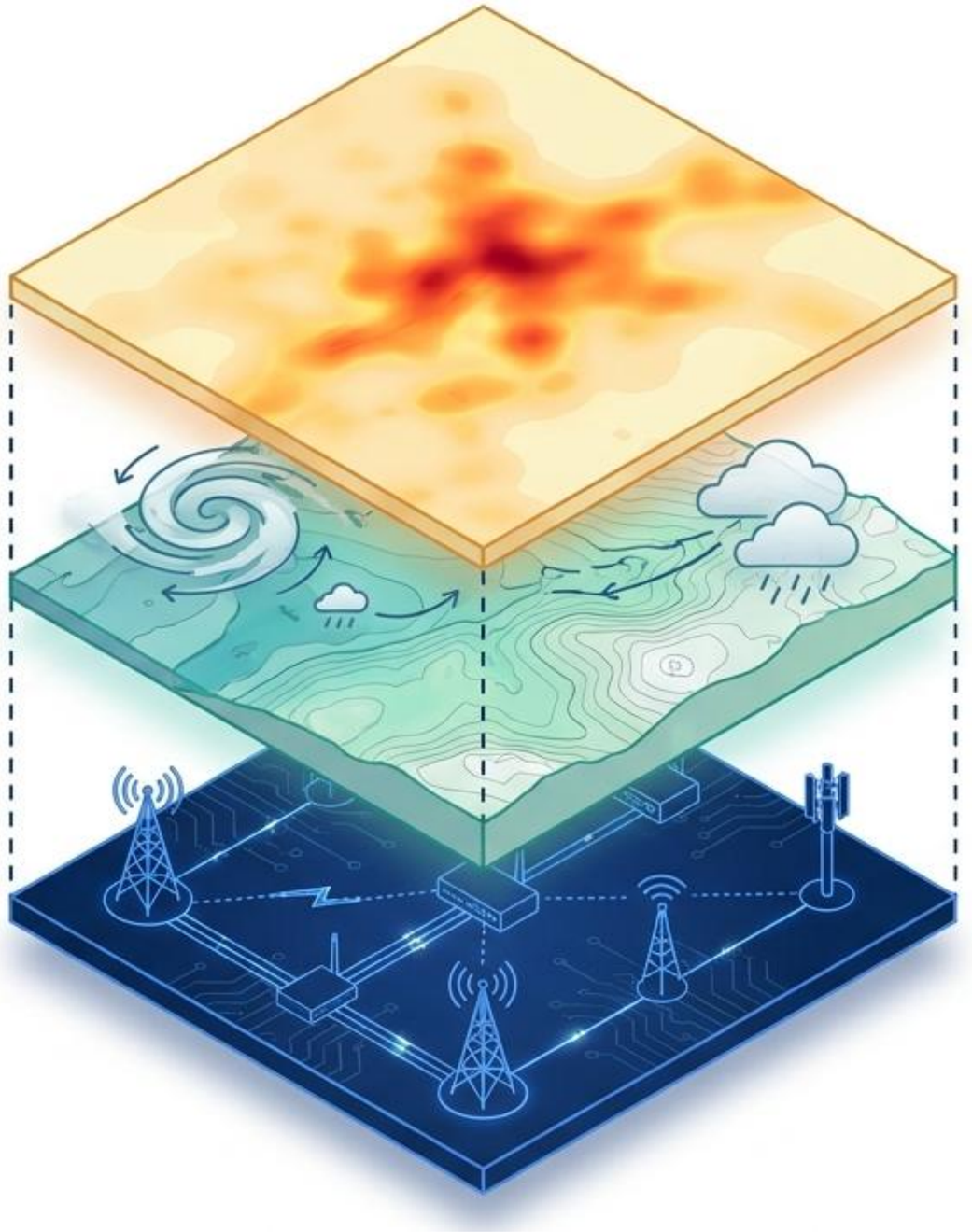
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Moving From Blind Networks to Context-Aware Infrastructure

Traditional network analysis operates in a vacuum, blind to the physical and human contexts that dictate true performance. Integrating Network KPIs with multi-sensor Geospatial Data (GeoAI) transforms telecom operations from reactive troubleshooting to predictive, intelligent planning.



Dr. Fathoni Usman, M.Eng., Eng.Tech., IPM, ASEAN Eng.
UNITEN

Raw Data Tables Cannot Reveal Geographic Blind Spots

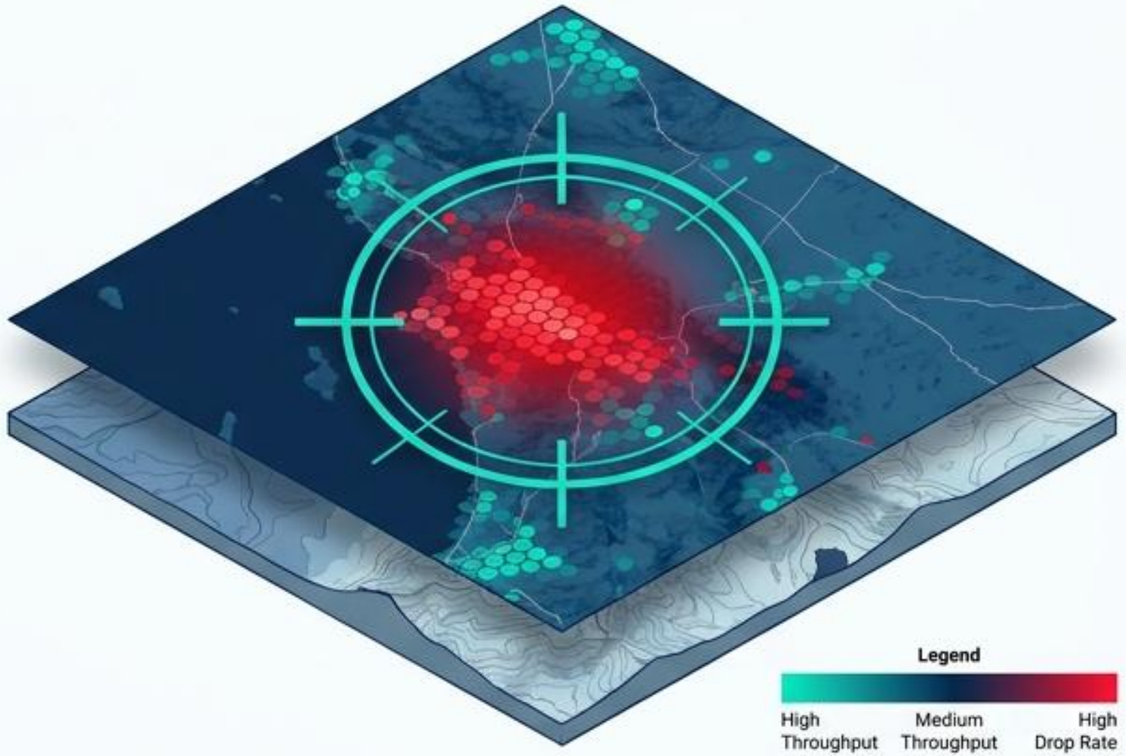
The Limitation

A spreadsheet of numbers forces engineers to react to symptoms without understanding the environment.

SITE_ID	LAT	LONG	THROUGHPUT	CALL_DROP_RATE
SITE_001	34.0522	-118.2437	120 Mbps	0.5%
SITE_002	34.0525	-118.2440	115 Mbps	0.7%
SITE_003	34.0530	-118.2430	125 Mbps	0.4%
SITE_004	34.0535	-118.2438	125 Mbps	0.5%
SITE_005	34.0530	-118.2480	125 Mbps	0.3%
SITE_015	34.0610	-118.2505	30 Mbps	5.2%
SITE_016	34.0610	-118.2570	25 Mbps	6.7%
SITE_027	34.0637	-118.2568	29 Mbps	5.9%
SITE_028	34.0701	-118.2601	25 Mbps	6.1%
SITE_030	34.0781	-118.2661	25 Mbps	6.7%
SITE_030	34.0781	-118.2480	25 Mbps	7.3%
SITE_042	34.0590	-118.2490	28 Mbps	5.8%
SITE_044	34.0590	-118.2490	28 Mbps	5.5%
SITE_086	34.0550	-118.2580	32 Mbps	4.7%
SITE_060	34.0715	-118.2580	32 Mbps	5.5%
SITE_067	34.0703	-118.2510	28 Mbps	6.7%
SITE_075	34.0585	-118.2510	29 Mbps	5.9%

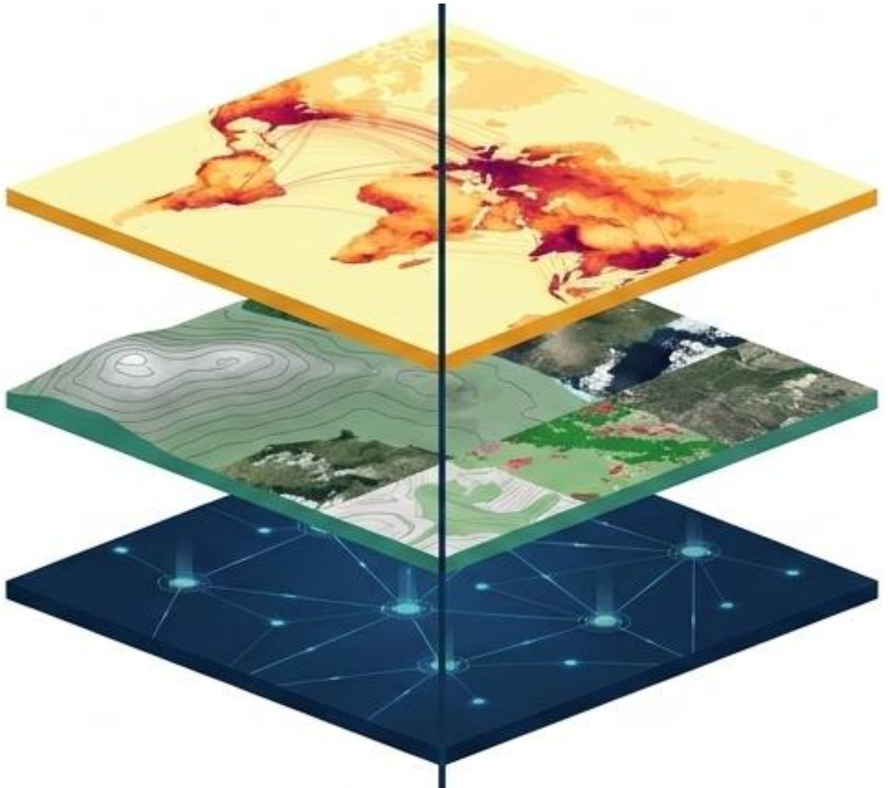
The Spatial Revelation

Plotting throughput, latency, and call-drop rates against geographic locations instantly shows where performance degrades. Engineers can visually correlate poor KPIs with terrain or interference in ways raw tables never reveal—prioritizing which sites need investigation first.



The Intelligence Stack: Fusing Three Dimensions of Reality

Layer 3: The Human Reality
Socio-Economic Data (Population density, Income, Mobility). The actual user demand.



Layer 2: The Physical Reality
Earth Observation (Terrain, Elevation, Weather, Land Cover). The environmental constraints.

Layer 1: The Technical Reality
Network KPIs (Throughput, Latency, Signal Strength). The network's own pulse.



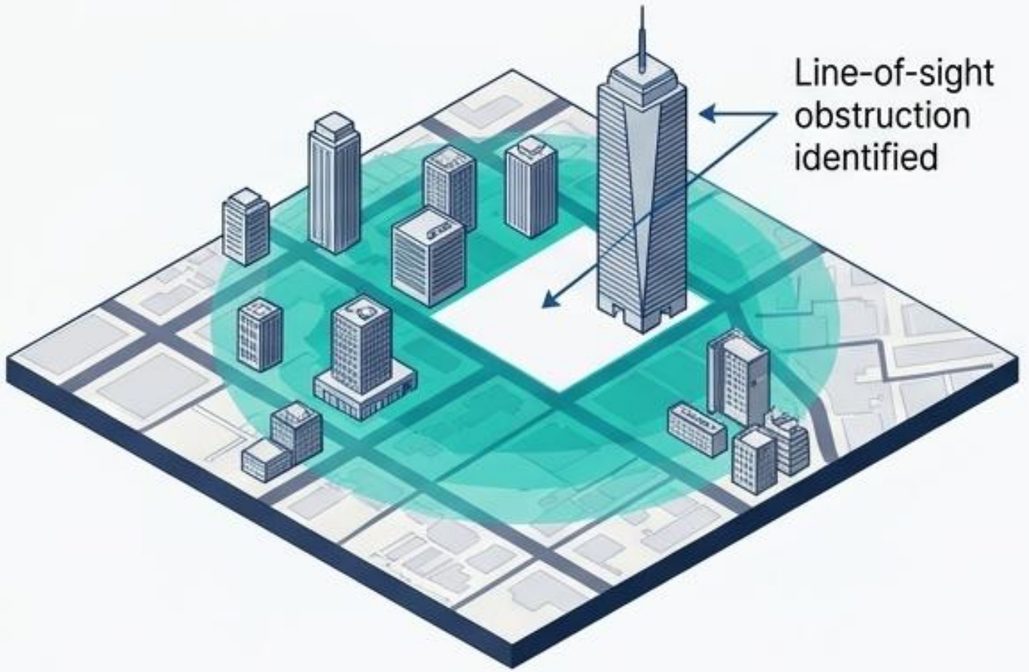
The Outcome
Actionable Decisions (Risk mitigation, resource optimization, predictive planning).

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Layer 1: Mapping the Network's True Coverage and Trends

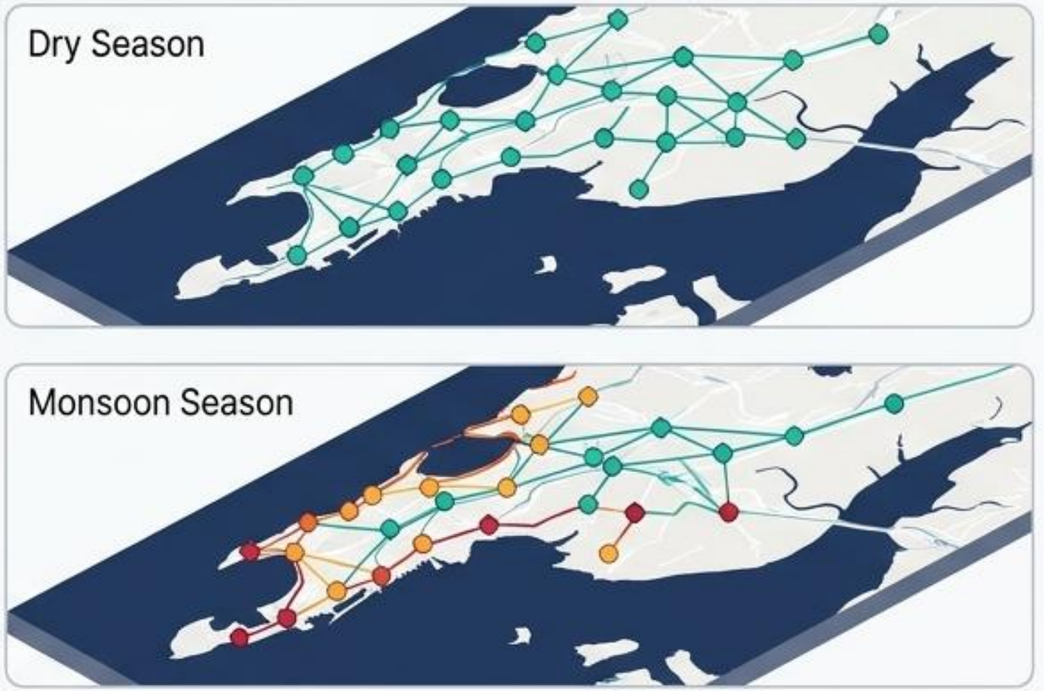
Coverage Visualization

Mapping signal strength exposes **marginal-coverage** pockets missed by customer complaints. Correlating dead zones with building data explains why a gap exists (line-of-sight obstruction vs. equipment fault), targeting the root cause.

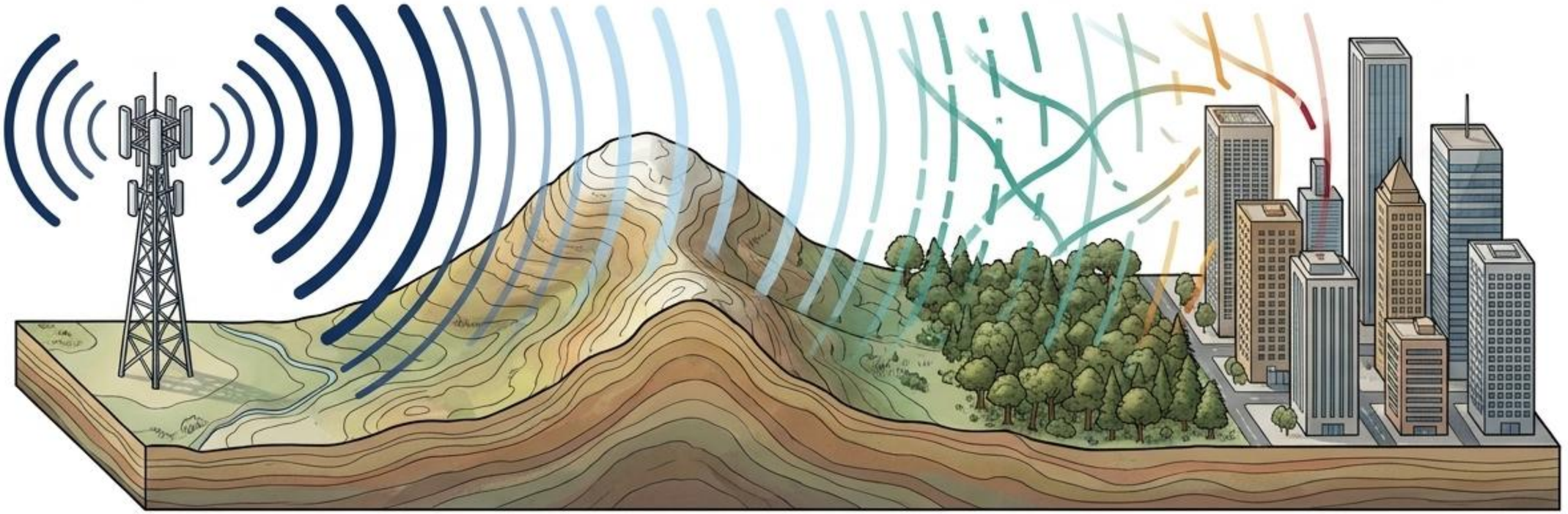


Spatially Aware Trend Analysis

Tracking KPI shifts across zones and seasons reveals hidden patterns. Longitudinal analysis (e.g., downtown congestion only during **commutes**, or **coastal degradation** during monsoons) allows operators to plan **capacity upgrades ahead of demand**.



Layer 2: The Physical Landscape Dictates Signal Propagation



Terrain and Land Cover

Hills, dense vegetation, and urban geometry block or attenuate radio signals differently across frequency bands.

Digital Elevation Models (DEM)

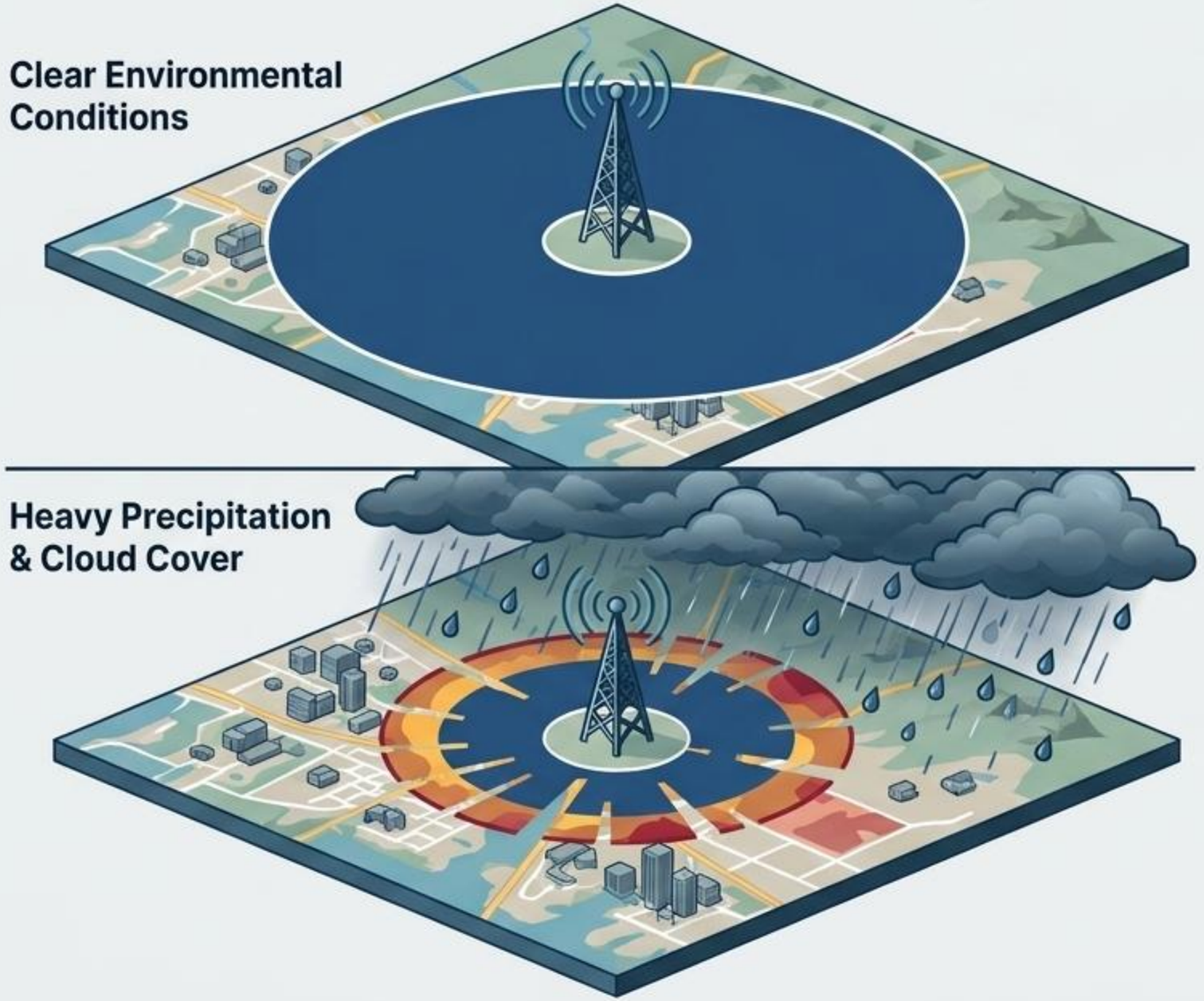
Utilizing Earth Observation data (like SRTM or Copernicus DEM) allows planners to simulate real-world coverage accurately.

The Value

Simulating signal propagation before a tower is built catches coverage problems on paper, avoiding expensive post-deployment failures.

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Transient Weather Data Explains Invisible Network Dips



The Weather Variable

Heavy rainfall, humidity, and storms noticeably degrade signal quality, especially at higher frequency bands used by modern technologies (like 5G).

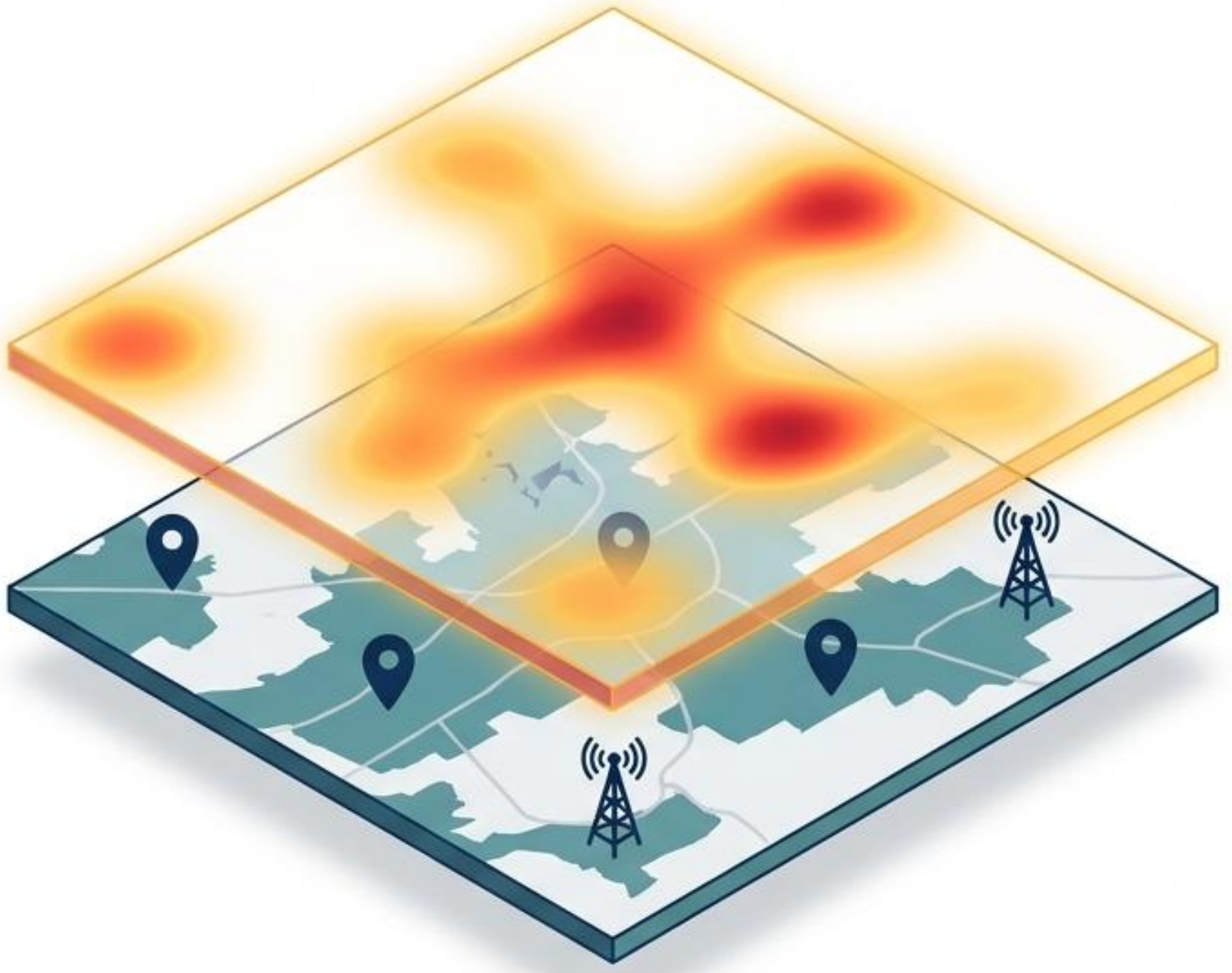
Diagnostic Clarity

By correlating historical climate products (like ERA5 or CHIRPS precipitation data) with network performance logs, operators can distinguish a genuine equipment hardware fault from a transient, weather-driven dip.

Action

Changes how—and how urgently—repair crews are deployed to reported outages.

Layer 3: Aligning Infrastructure with Human Reality



Demand Overlay

Following the User

Capacity planning must follow where people actually are, not just where towers currently exist.

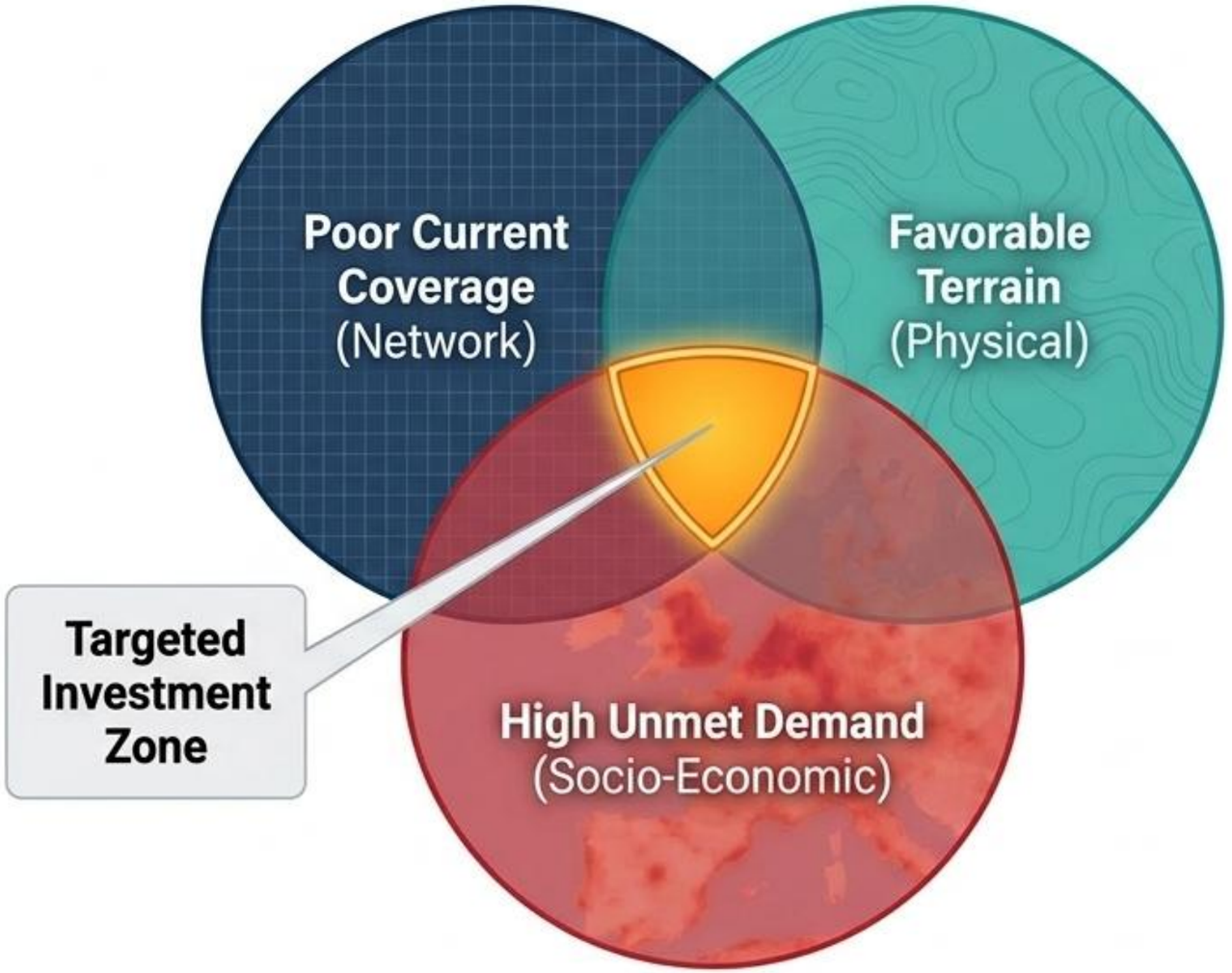
Predicting Congestion

Fusing high-resolution population grids and mobility data with network KPIs identifies high-density areas approaching capacity limits before users experience congestion.

Contextual Pricing

Understanding local income levels and typical data consumption patterns helps operators forecast realistic demand and appropriately price services, avoiding a one-size-fits-all rollout failure.

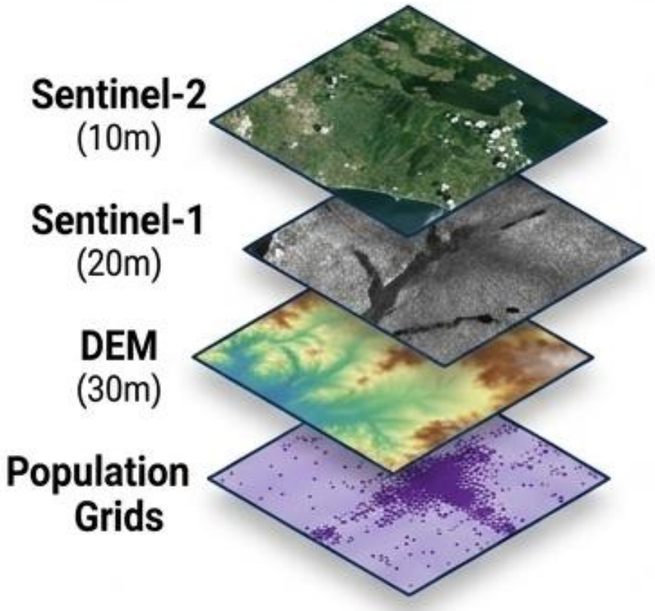
Identifying the Underserved Gaps for Targeted Investment



- **Revealing the Gaps:** Overlaying coverage maps with socio-economic indicators instantly surfaces communities that are both poorly served and possess genuine unmet demand.
- **Maximum Return:** These specific intersections represent the areas where targeted capital investment delivers the highest possible commercial and social return.
- **Regulatory Alignment:** This socio-economic fusion increasingly guides government universal-service funding and policy prioritization.

The Engineering Reality: Harmonizing the Data Stack

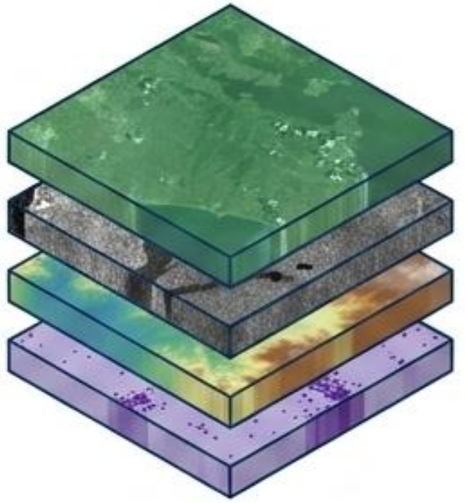
Stage 1: Input



Stage 2: Harmonization

- Resampling
- Reprojection
- Alignment

Stage 3: Output



Unified Multi-Layer Data Cube
(Same projection, resolution, time frame)

Geospatial fusion fails if the layers don't align. Different sensors mean different resolutions, coordinate systems, and formats.

Data Harmonization
The essential process of making diverse datasets (optical, SAR, climate, population) compatible.

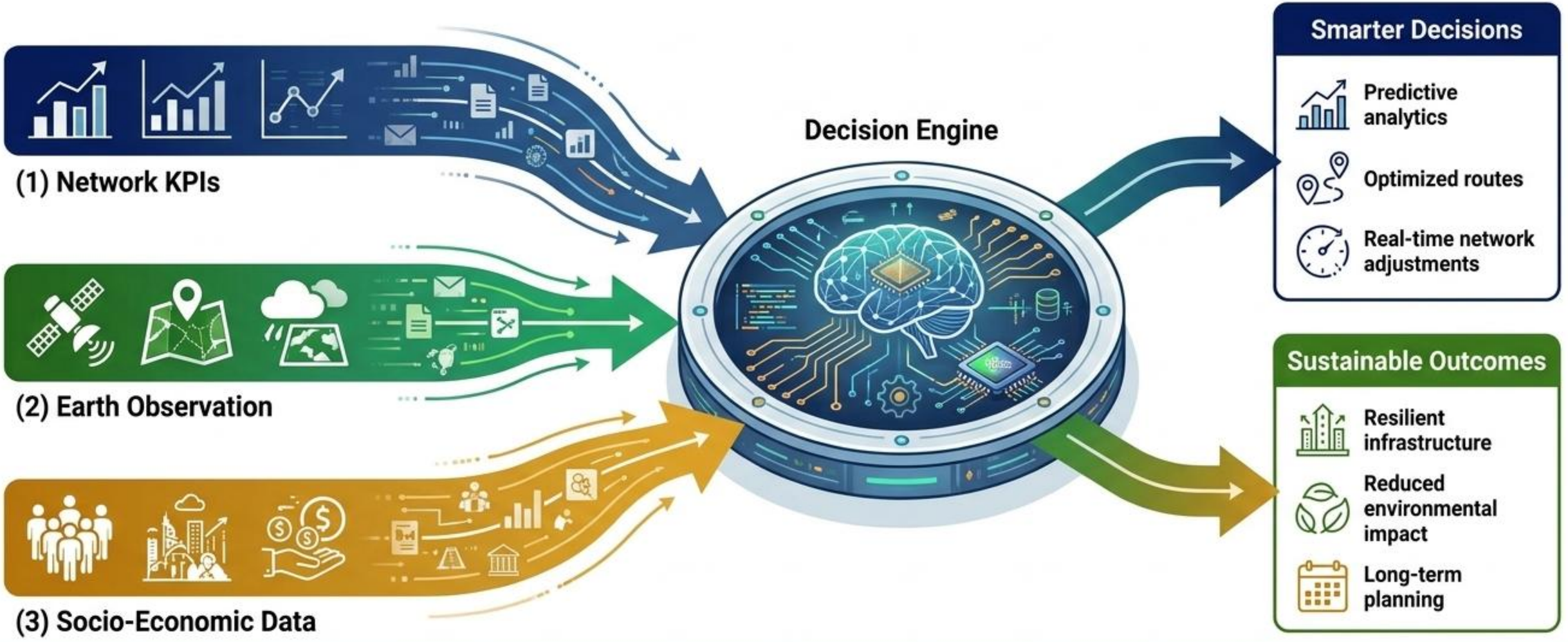
The AI-Ready Cube
Consistent scales, unified Coordinate Reference Systems (CRS), and precise cloud masking create the foundational 4D data cube required for reliable GeoAI models.

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The Network Planning Evolution: A Comparative Diagnostic

	Siloed Network Operations	GeoAI-Fused Operations
Dimension 1: Troubleshooting	Reactive to customer complaints and arbitrary KPI drops.	Spatial KPI visualization pinpoints environmental vs. hardware root causes instantly.
Dimension 2: Capacity Planning	Tower-centric forecasting based solely on past utilization.	Socio-economic demand forecasting aligned with actual human mobility and income.
Dimension 3: Maintenance	Post-event repair after infrastructure is damaged.	Predictive response utilizing Earth observation (e.g., weather/flood data) to pre-position crews.

The Unified Geospatial Telecom Framework

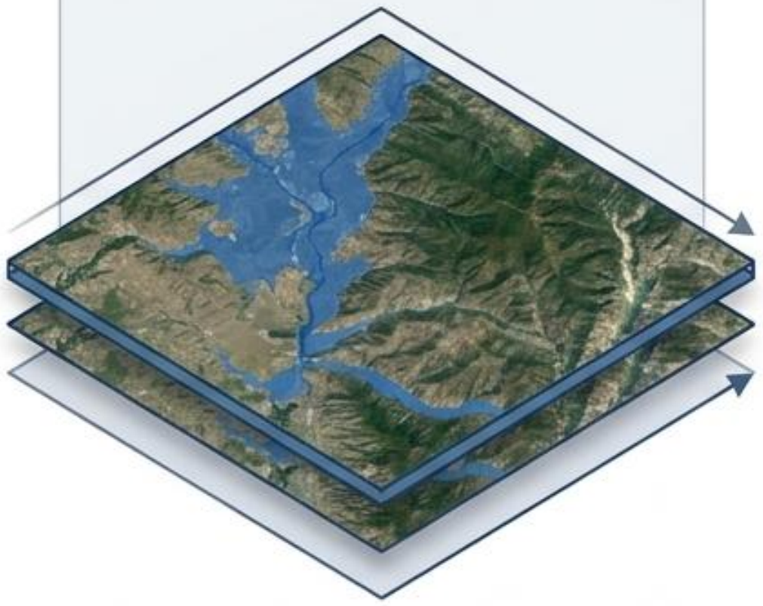


Stronger Together: True network intelligence does not emerge from individual datasets, but exclusively from their intersection. Multi-sensor fusion delivers deeper, more reliable, and actionable geospatial intelligence. It bridges the gap between the technical metrics of the machine and the physical reality of the user.

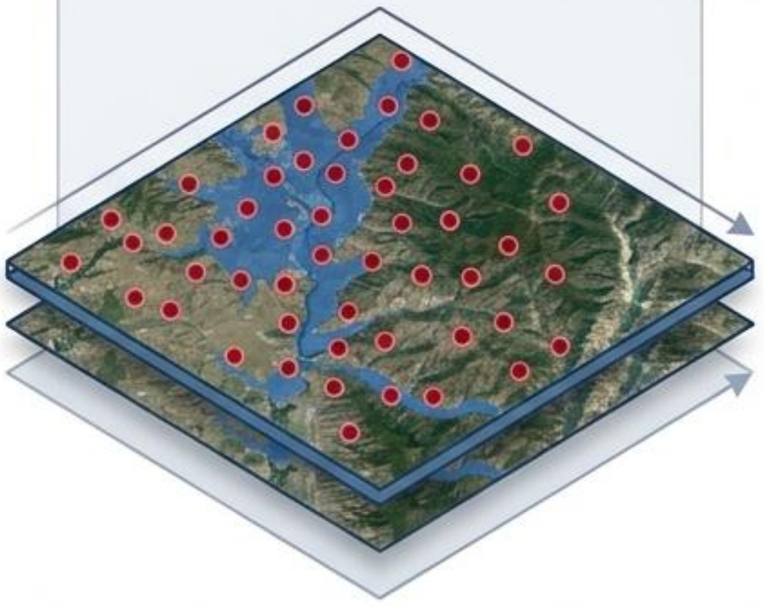
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Real-World Fusion: Flood-Aware Site Planning

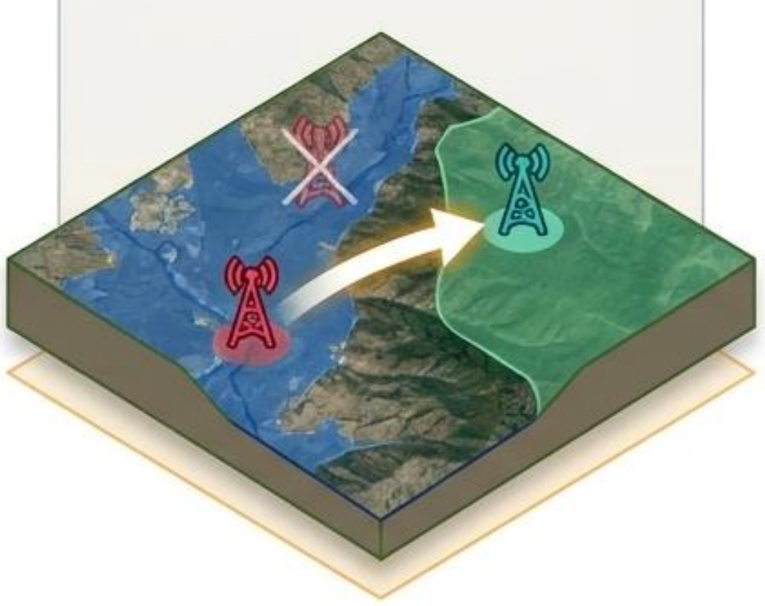
Input Data



Spatial Analysis



Operational Action



The Risk

Siting critical infrastructure in unmapped, seasonal flood zones guarantees future outages and massive repair bills.

The Preventive Fix

By overlaying historical flood-extent data derived from EO imagery onto proposed sites, planners can adjust locations before concrete is poured.

The ROI

Preventive spatial analysis is exponentially cheaper than the cost of relocating or repairing a water-damaged site.

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Real-World Fusion: Predictive Outage Response

Shifting to Predictive Maintenance

Fusing real-time weather feeds with historic network performance data enables operators to anticipate exact outage locations.

Pre-Positioning

Instead of waiting for customer complaints, dispatchers can pre-position repair crews in high-risk zones ahead of extreme weather events.

The Impact

Measurably reduces both network downtime and the volume of customer-reported complaints, protecting brand reputation.



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Architecting the Next Generation of ASEAN Infrastructure

- The future of telecommunications requires seeing the network not as a closed circuit, but as a living system embedded in the physical and human world.
- Mastering Geospatial Data Engineering and Feature Extraction is the critical step toward building resilient, profitable, and equitable networks.

